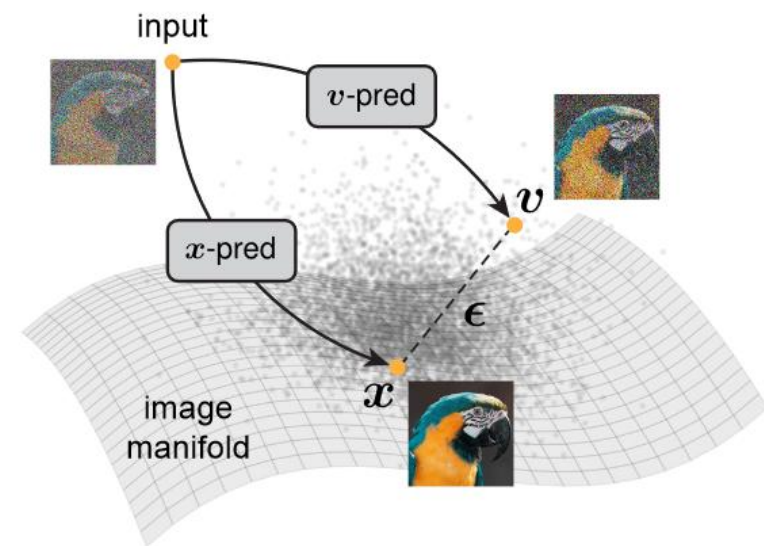
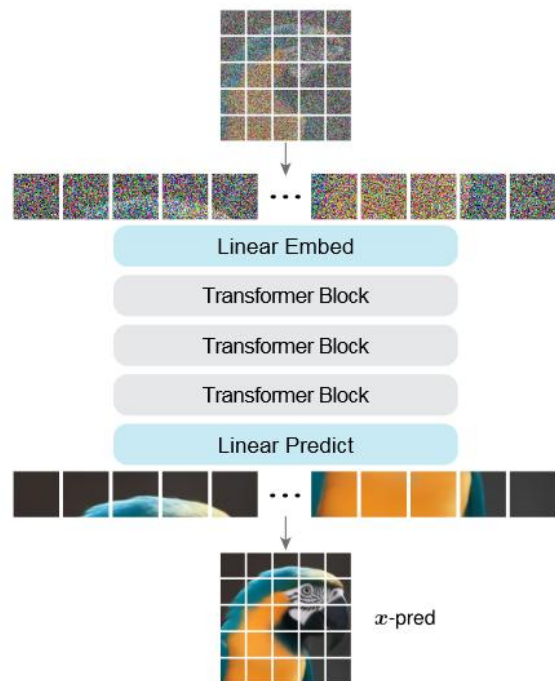


Back to Basics: Let Denoising Generative Models Denoise

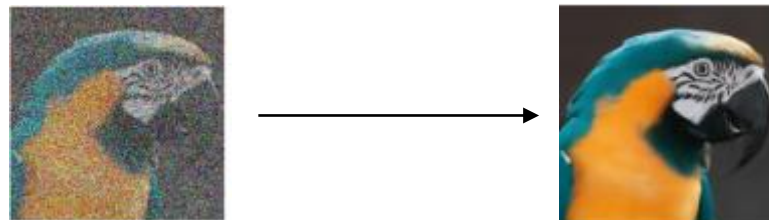
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MIT



Diffusion Model

初衷：损坏图像 $\xrightarrow[\text{denoise}]{\text{去噪}}$ 干净图像



DDPM

$$\mu_{\theta}(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \alpha_t}} \epsilon_{\theta}(x_t, t) \right)$$

ϵ -prediction

Flow Matching

$$\frac{d\phi_t(x)}{dt} = v_{\theta}(\phi_t(x))$$

v -prediction

Diffusion Model 的发展

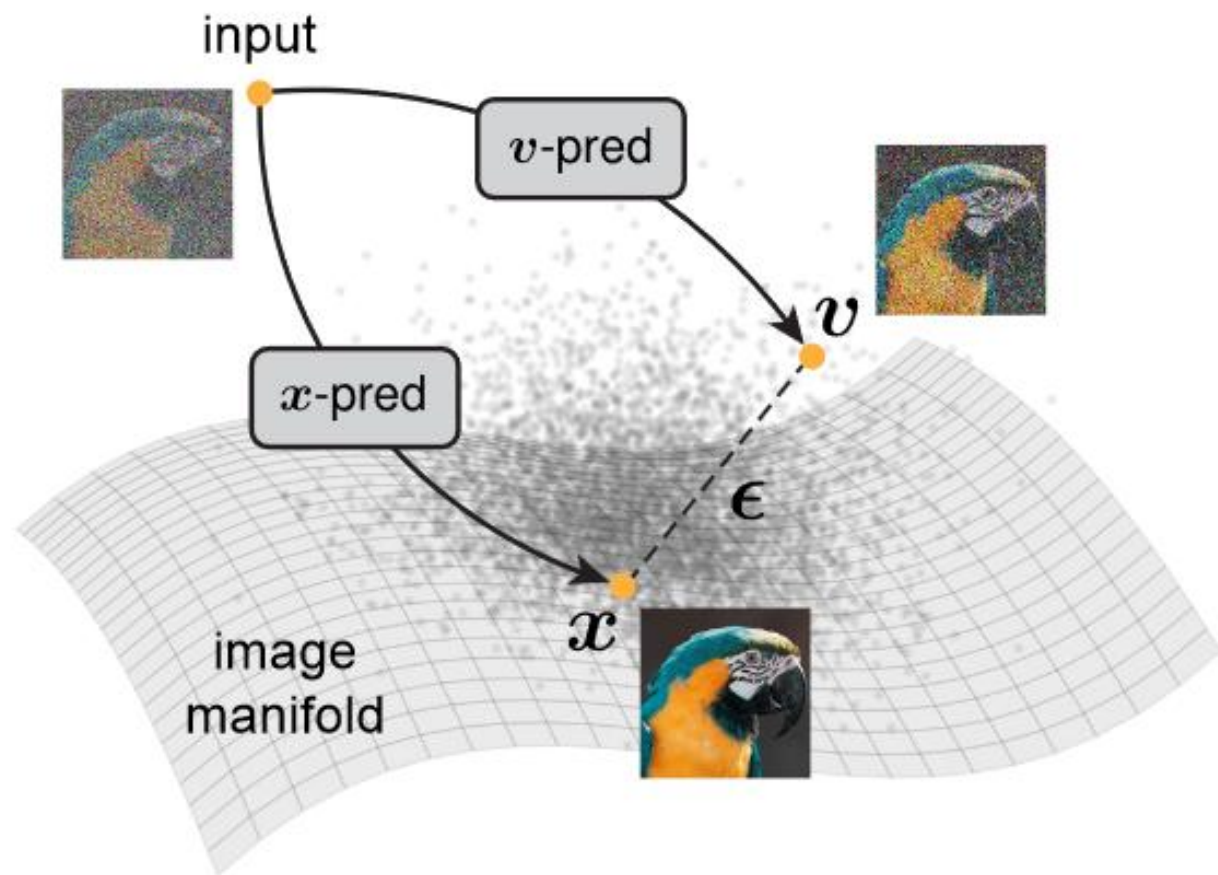
不直接去噪，而是预测含噪量，背离了直接预测干净图像的初衷

Manifold Assumption——流形假设

- 自然数据位于低维流形
- 噪声、流速位于高维流形

直观上看：

- 预测 x →只需拟合低维信息
- 预测 ϵ, v →需拟合高维信息



Problem Setting

流的角度 $x \sim p_{data}(x)$, $\epsilon \sim \mathcal{N}(0, I)$ $\longrightarrow$ $z_t = tx + (1 - t)\epsilon$

$v = z'_t = x - \epsilon$ $\longrightarrow$ $L = E_{t,x,\epsilon} \|v_\theta(z_t, t) - v\|^2$ $\longrightarrow$ ODE数值迭代

$$\begin{cases} z_t = tx + (1 - t)\epsilon \\ v = z'_t = x - \epsilon \end{cases} \implies \begin{cases} x = (1 - t)v + z_t \\ \epsilon = z_t - tv \end{cases}$$

九种组合（3种网络预测*3种损失函数）

	(a) x -pred $x_\theta := \text{net}_\theta(z_t, t)$	(b) ϵ -pred $\epsilon_\theta := \text{net}_\theta(z_t, t)$	(c) v -pred $v_\theta := \text{net}_\theta(z_t, t)$
(1) x -loss: $\mathbb{E} \ x_\theta - x\ ^2$	x_θ	$x_\theta = (z_t - (1 - t)\epsilon_\theta)/t$	$x_\theta = (1 - t)v_\theta + z_t$
(2) ϵ -loss: $\mathbb{E} \ \epsilon_\theta - \epsilon\ ^2$	$\epsilon_\theta = (z_t - tx_\theta)/(1 - t)$	ϵ_θ	$\epsilon_\theta = z_t - tv_\theta$
(3) v -loss: $\mathbb{E} \ v_\theta - v\ ^2$	$v_\theta = (x_\theta - z_t)/(1 - t)$	$v_\theta = (z_t - \epsilon_\theta)/t$	v_θ

Toy Experiment

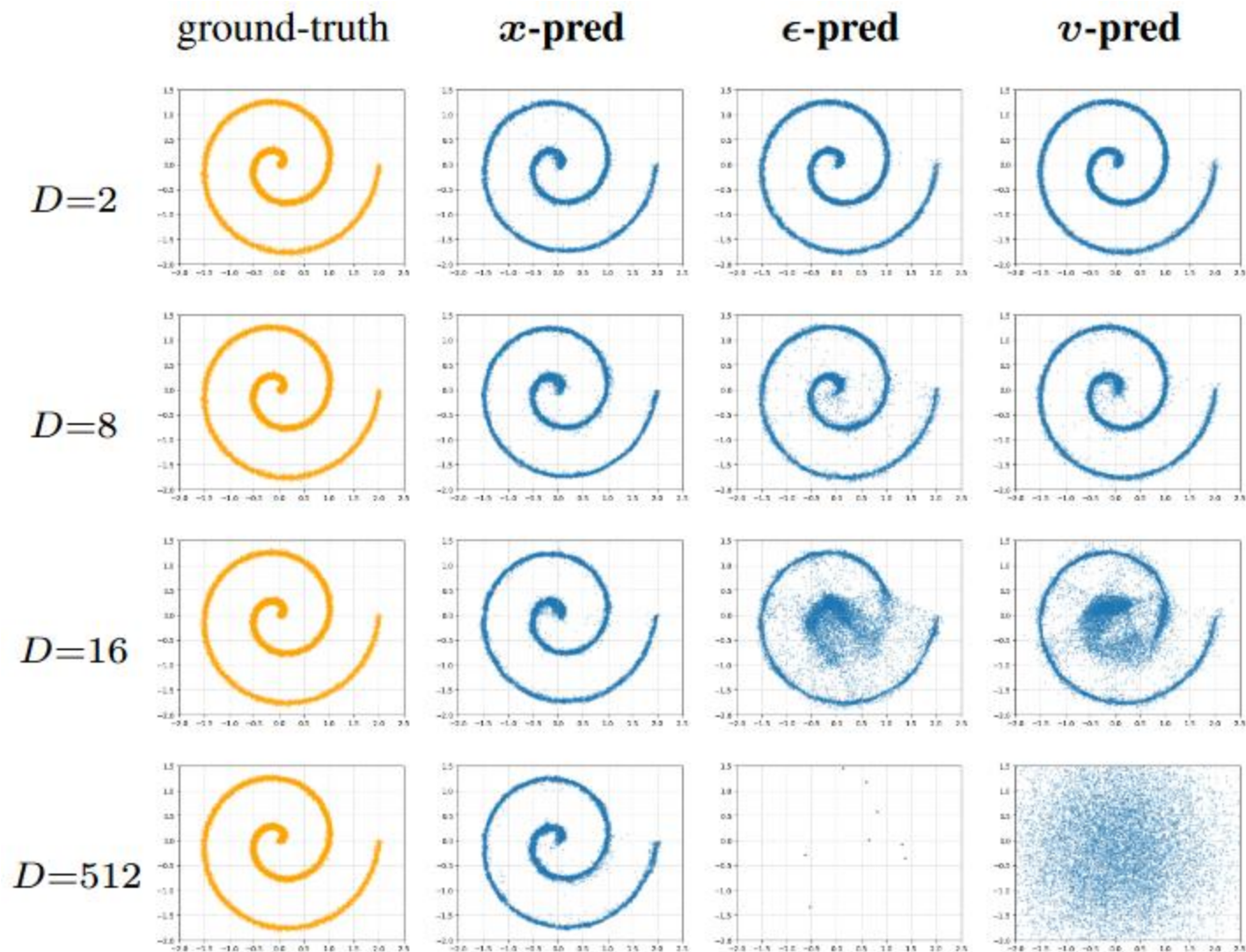
$\mathbf{x}$ -prediction的本质优势：只学低维流形，低容量网络也能做（ $\epsilon, \mathbf{v}$ -prediction原则上需要高容量）

玩具实验的setting

低维 ($d=2$) 数据嵌入高维 D

D 维生成任务

统一用256维的5层MLP训练网络

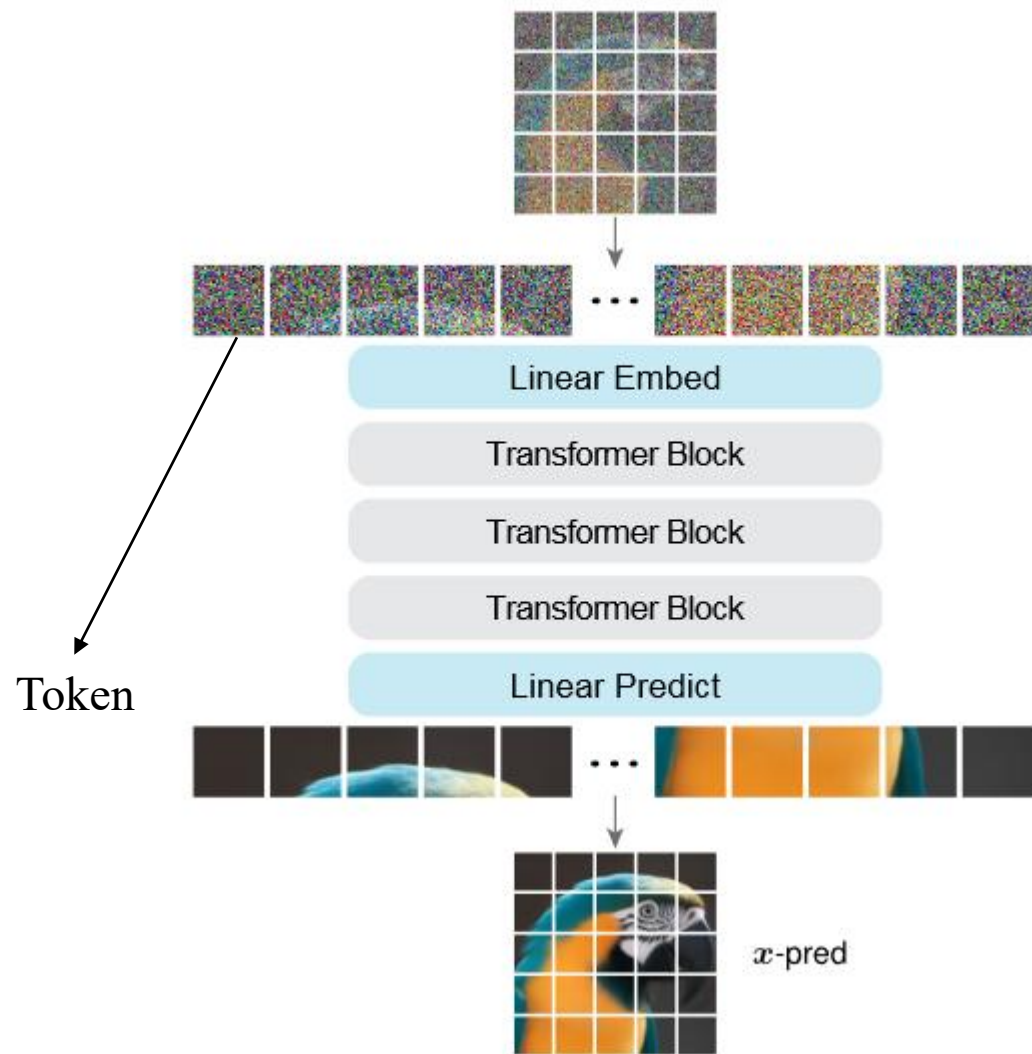


Just Image Transformers(JiT)

图片分块的Transformer

$$H \times W \times C \text{ 图像} \longrightarrow \frac{H}{p} \times \frac{W}{p} \times C \text{ patch}$$

极简主义的设计：无分类器、预训练、额外损失



Experiments

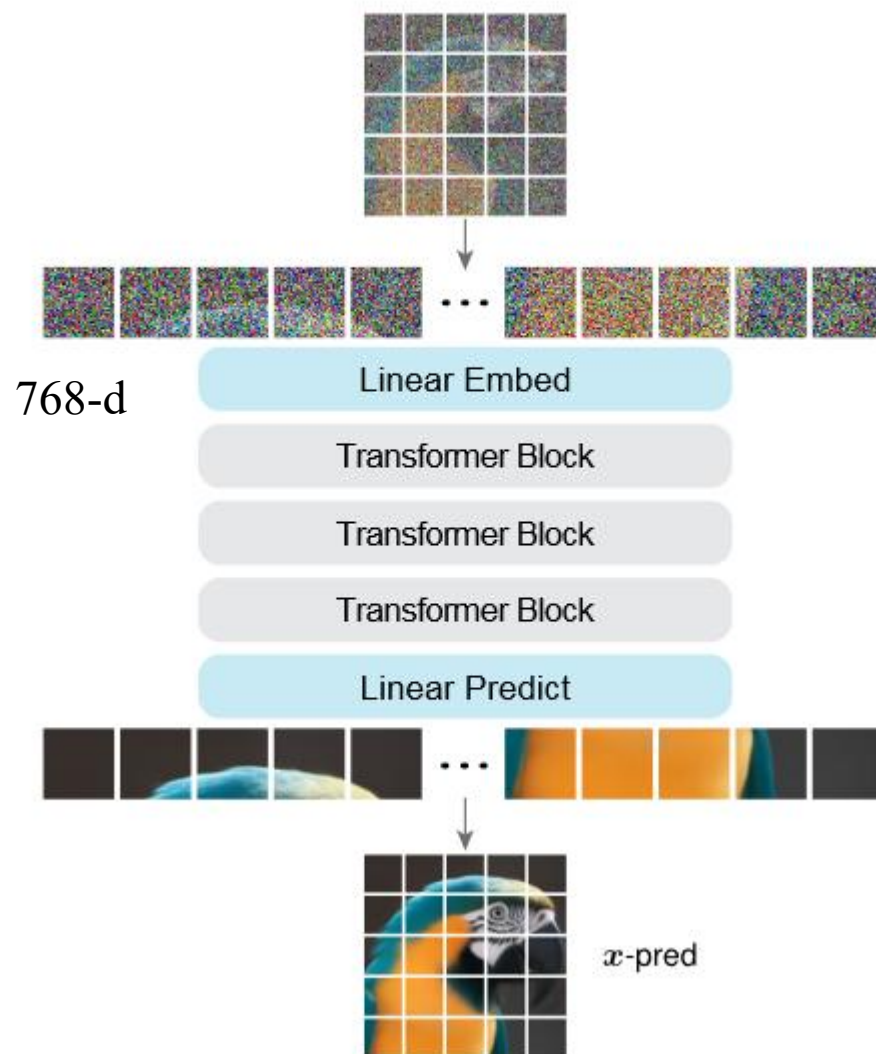
	x -pred	ϵ -pred	v -pred
x -loss	10.14	379.21	107.55
ϵ -loss	10.45	394.58	126.88
v -loss	8.62	372.38	96.53

(a) ImageNet 256×256 , JiT-B/16

	x -pred	ϵ -pred	v -pred
x -loss	5.76	6.20	6.12
ϵ -loss	3.56	4.02	3.76
v -loss	3.55	3.63	3.46

(b) ImageNet 64×64 , JiT-B/4

x -prediction is critical



Experiments

t -shift (μ)		x -pred	ϵ -pred	v -pred
(lower noise)	0.0	14.44	464.25	120.03
	-0.4	9.79	372.91	109.93
	-0.8	8.62	372.36	96.53
(higher noise)	-1.2	8.99	355.25	106.85

$$t \in (0,1)$$

$$\text{logit}(t) = \ln \frac{t}{1-t} \sim \mathcal{N}(\mu, \sigma^2)$$

噪声水平提高对高分辨率图像扩散有效，但不能根本解决问题

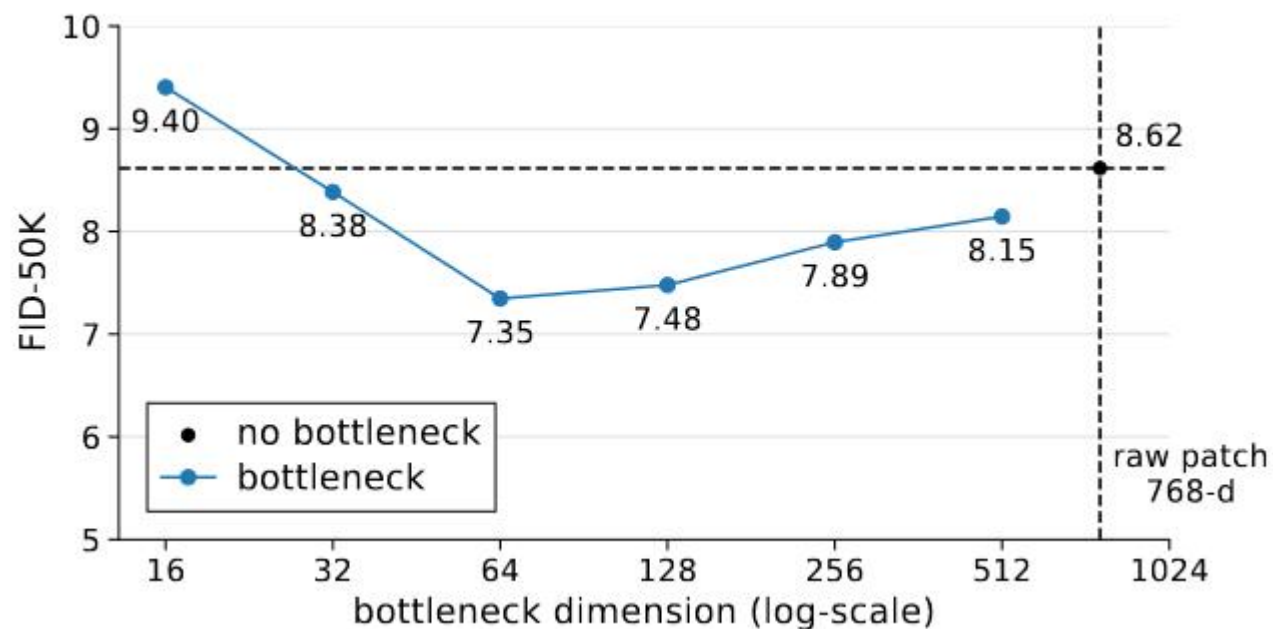
本质还是有限的网络大小无法拟合高维信息

Noise-level shift is not sufficient

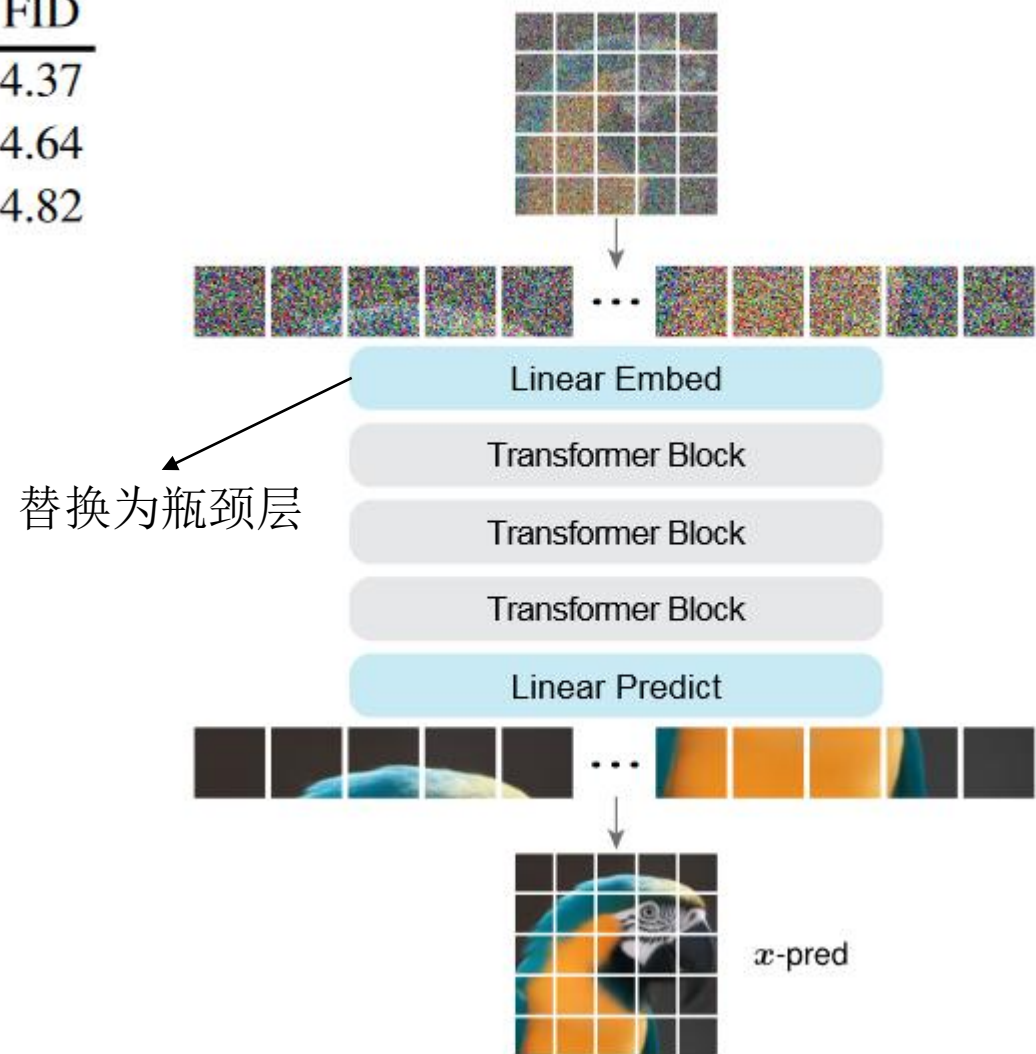
Experiments

resolution	model	len	patch dim	hiddens	params	Gflops	FID
256×256	JiT-B/16	256	768	768	131	25	4.37
512×512	JiT-B/32	256	3072	768	133	26	4.64
1024×1024	JiT-B/64	256	12288	768	141	30	4.82

Increasing hidden units is not necessary



Bottleneck can be beneficial



Experiments

	JiT-B/16	JiT-L/16
Baseline (SwiGLU, RMSNorm)	7.48 (6.32)	-
+ RoPE, qk-norm	6.69 (5.44)	-
+ in-context class tokens	5.49 (4.37)	3.39 (2.79)

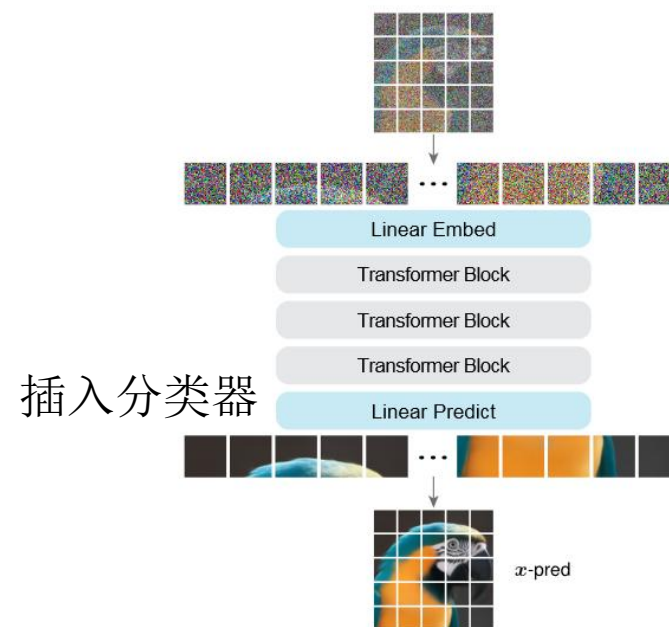
引入Transformer的通用改进SwiGLU、RMSNorm、RoPE、qk-norm

256×256	200-ep	600-ep	512×512	200-ep	600-ep
JiT-B/16	4.37	3.66	JiT-B/32	4.64	4.02
JiT-L/16	2.79	2.36	JiT-L/32	3.06	2.53
JiT-H/16	2.29	1.86	JiT-H/32	2.51	1.94
JiT-G/16	2.15	1.82	JiT-G/32	2.11	1.78

不同size的模型，可扩展性（scaling）

FID-50K	JiT-B/16	JiT-L/16
<i>v</i> -loss only	4.37	2.79
w/ cls loss	4.14	2.50

引入分类器的loss



Experiments

对比实验

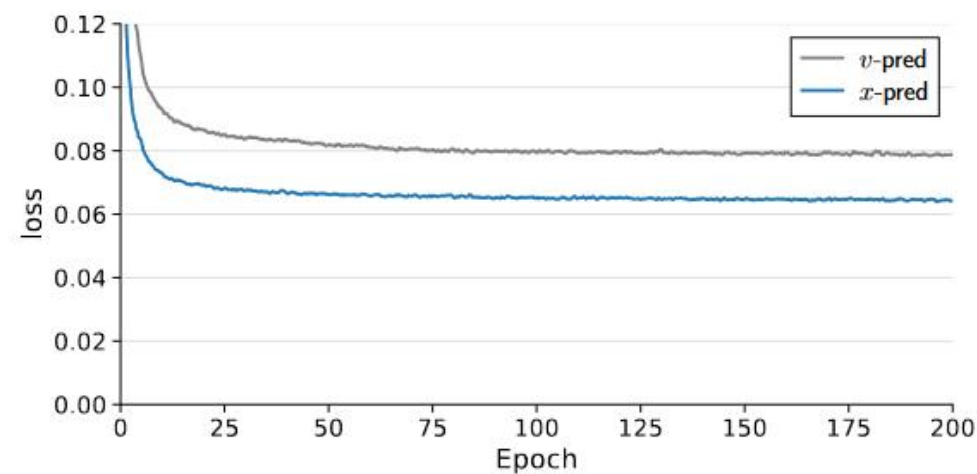
无需预训练 额外损失 轻量

ImgNet 256×256	pre-training			params	Gflops	FID↓	IS↑
	token	perc.	self-sup.				
<i>Latent-space Diffusion</i>							
DiT-XL/2 [46]	SD-VAE	VGG	-	675+49M	119	2.27	278.2
SiT-XL/2 [40]	SD-VAE	VGG	-	675+49M	119	2.06	277.5
REPA [74], SiT-XL/2	SD-VAE	VGG	DINOv2	675+49M	119	1.42	305.7
LightningDiT-XL/2 [73]	VA-VAE	VGG	DINOv2	675+49M	119	1.35	295.3
DDT-XL/2 [71]	SD-VAE	VGG	DINOv2	675+49M	119	1.26	310.6
RAE [78], DiT ^{DH} -XL/2	RAE	VGG	DINOv2	839+415M	146	1.13	262.6
<i>Pixel-space (non-diffusion)</i>							
JetFormer [65]	-	-	-	2.8B	-	6.64	-
FractalMAR-H [36]	-	-	-	848M	-	6.15	348.9
<i>Pixel-space Diffusion</i>							
ADM-G [12]	-	-	-	554M	1120	4.59	186.7
RIN [28]	-	-	-	410M	334	3.42	182.0
SiD [25] , UViT/2	-	-	-	2B	555	2.44	256.3
VDM++, UViT/2	-	-	-	2B	555	2.12	267.7
SiD2 [26], UViT/2	-	-	-	N/A	137	1.73	-
SiD2 [26], UViT/1	-	-	-	N/A	653	1.38	-
PixelFlow [6], XL/4	-	-	-	677M	2909	1.98	282.1
PixNerd [70], XL/16	-	-	DINOv2	700M	134	2.15	297
JiT-B/16	-	-	-	131M	25	3.66	275.1
JiT-L/16	-	-	-	459M	88	2.36	298.5
JiT-H/16	-	-	-	953M	182	1.86	303.4
JiT-G/16	-	-	-	2B	383	1.82	292.6

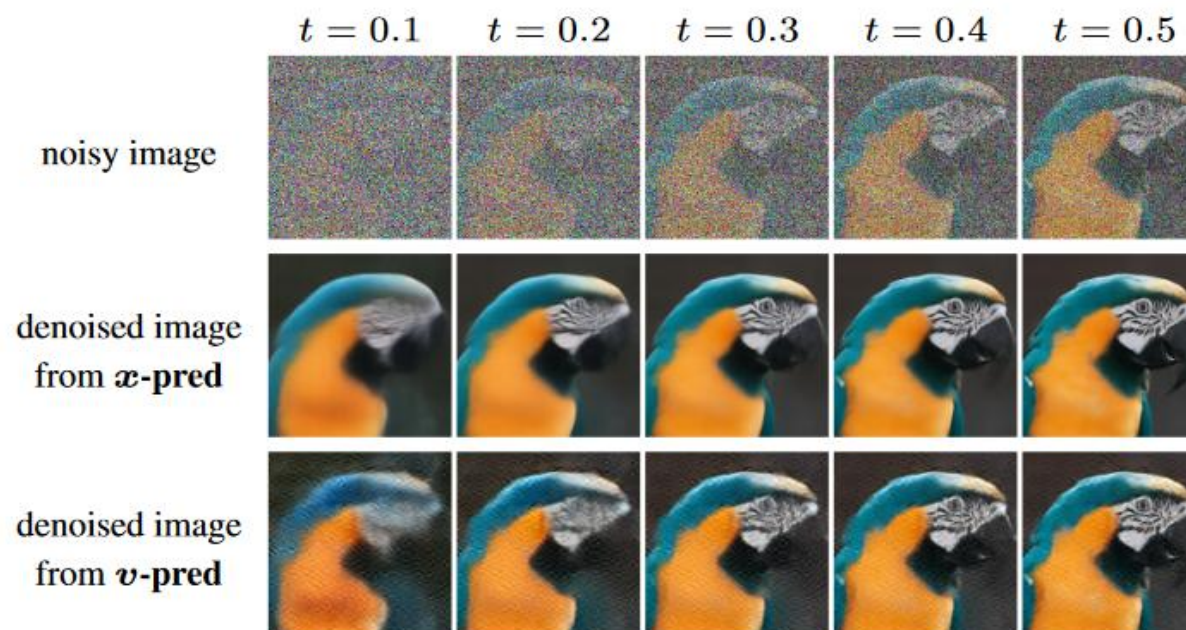
Additional Experiments

$$L = E_{t,x,\epsilon} ||v_{\theta}(z_t, t) - v||^2$$

v -prediction loss 更高，无法承担学习高维特征任务



v -prediction 图像伪影



Additional Experiments

EDM $x_\theta(z_t, t) = c_{skip}z_t + c_{out}net_\theta(z_t, t)$ 让训练更稳定

EDM不是 x -prediction

在高维像素空间EDM失效

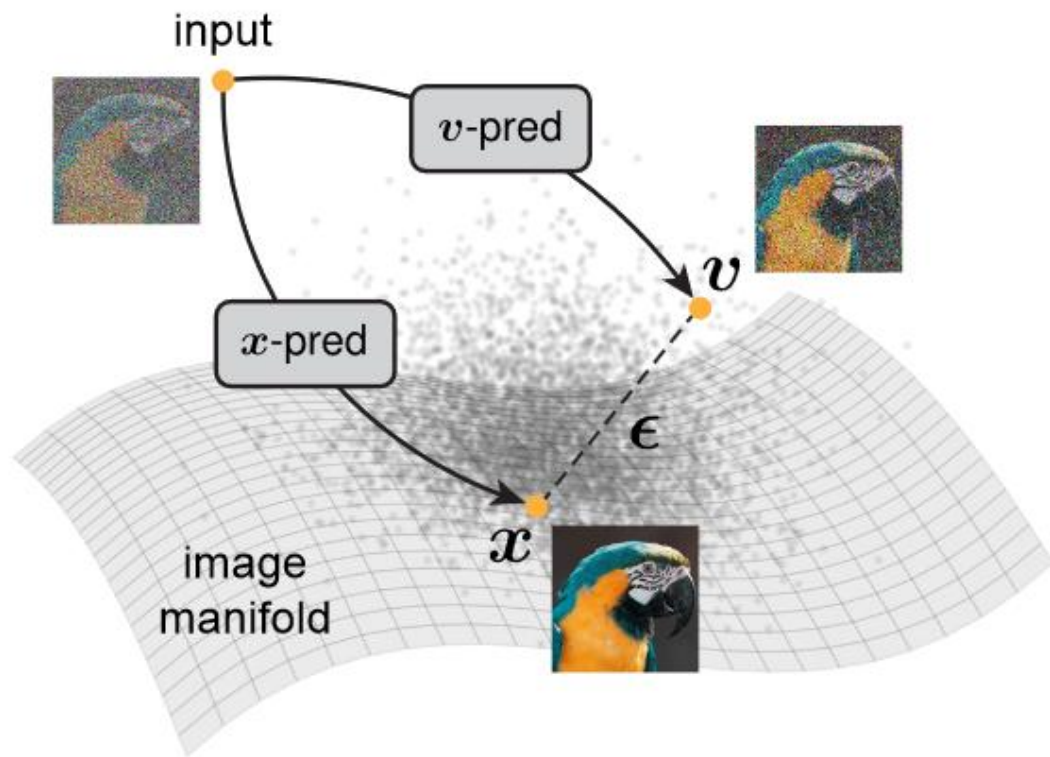
	x -pred	pre-conditioned predictions	
		EDM-style	linear
	$c_{skip} = 0$ $c_{out} = 1$	$c_{skip} = \frac{1}{t} \frac{\sigma_{data}^2}{\sigma_{data}^2 + \sigma_t^2}$ $c_{out} = \frac{\sigma_{data} \sigma_t}{\sqrt{\sigma_{data}^2 + \sigma_t^2}}$	$c_{skip} = t$ $c_{out} = 1 - t$
x -loss	10.14	28.94	39.50
ϵ -loss	10.45	72.05	67.56
v -loss	8.62	35.49	46.25

	FID@256
JiT-G/16@256	1.82
JiT-G/32@512, ↓256	1.84

	FID@512
JiT-G/16@256, ↑512	2.45
JiT-G/32@512	1.78

跨分辨率生成

流形假设



x -prediction

Transformer + 流形假设+小容量网络

x -prediction

在DDPM失败 (U-net) 不同t的损失尺度差别大

x -prediction

